17th National Undergraduate Mathematics Olympiad 2026

80 marks | 120 Minutes

(re-constructed)

Rules:

- No Electronics allowed (Phone / Calculator)
- ...

Q1. Show that 1^241 + 2^241 + 3^241 + 4^241 is divisible by 5 but 1^240 + 2^240 + 3^240 + 4^240 is not divisible by 5.

Q2. (incomplete) Working with λ and eigenvalues... matrix math (n x n)... A^3 = A^T ... find value of A (= 2 or 4)

Q3. Fibonacci series Fn with F1 = 1, F2 = 2, F(n+1) = Fn + F(n-1)
Find

$$\sum_{K=1}^{\infty} \frac{F_K}{3^K}$$

Q4. (incomplete) Domain-Range math and finding Riemann Integrable

given $f : [0,1] \to \mathbb{R}$

$$f(x) = \begin{cases} \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ is a rational number in lowest terms,} \\ 0, & \text{if } x \text{ is an irrational number in } (0,1). \end{cases}$$

Where $p \in \mathbb{Z}, q \in \mathbb{N}, q \geq 1, \text{ and } \gcd(p,q) = 1$

Q5. (Limit Math) Given f'(x) and g'(x) exists. g'(x) ≠ 0 and x ∈ (to - δ, to + δ).
given,

$$\lim_{t \to t_0} \frac{f'(x)}{g'(x)} = K$$

prove,

$$\lim_{t \to t_0} \frac{f(x)}{g(x)} = k$$

Q6. (incomplete) For a recurrence series x(n), x1 = 1/2,

$$x_{n+1} = \frac{2x_n}{1+x_n^2}$$

Define S(n) and it's given form (similar to)

$$S_n = \sum \frac{x_n'}{1+x_n^{2'}}$$

if x1 = t then find explicitly S(n) in terms of n, x, t

Q7. (incomplete) Domino (1 x 2) and Rectangular Board (2 x n) math. How many ways Tn can the Domino tiles be placed on the Rectangular board in vertical / horizontal fashion? find: (i) Recurrence function (ii) Closed Form expression (iii) generating function

$$\sum_{k=0}^{n} T_n x^n$$

Q8. (incomplete) given,

$$\theta = \frac{2\pi}{n}$$

where n >= 2 and points P(0,k), rotation on plane-XY as T(k). Rotating counterclockwise in T1, T2, T3 and find composition as R(x,y) in n, k, (?)

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for more questions, visit: https://b1tacad.netlify.app/#Qbank

you can write your draft and use markdown / html syntax to convert it into PDF using tools like these: https://b1tranger.netlify.app/#pdf-make-tool